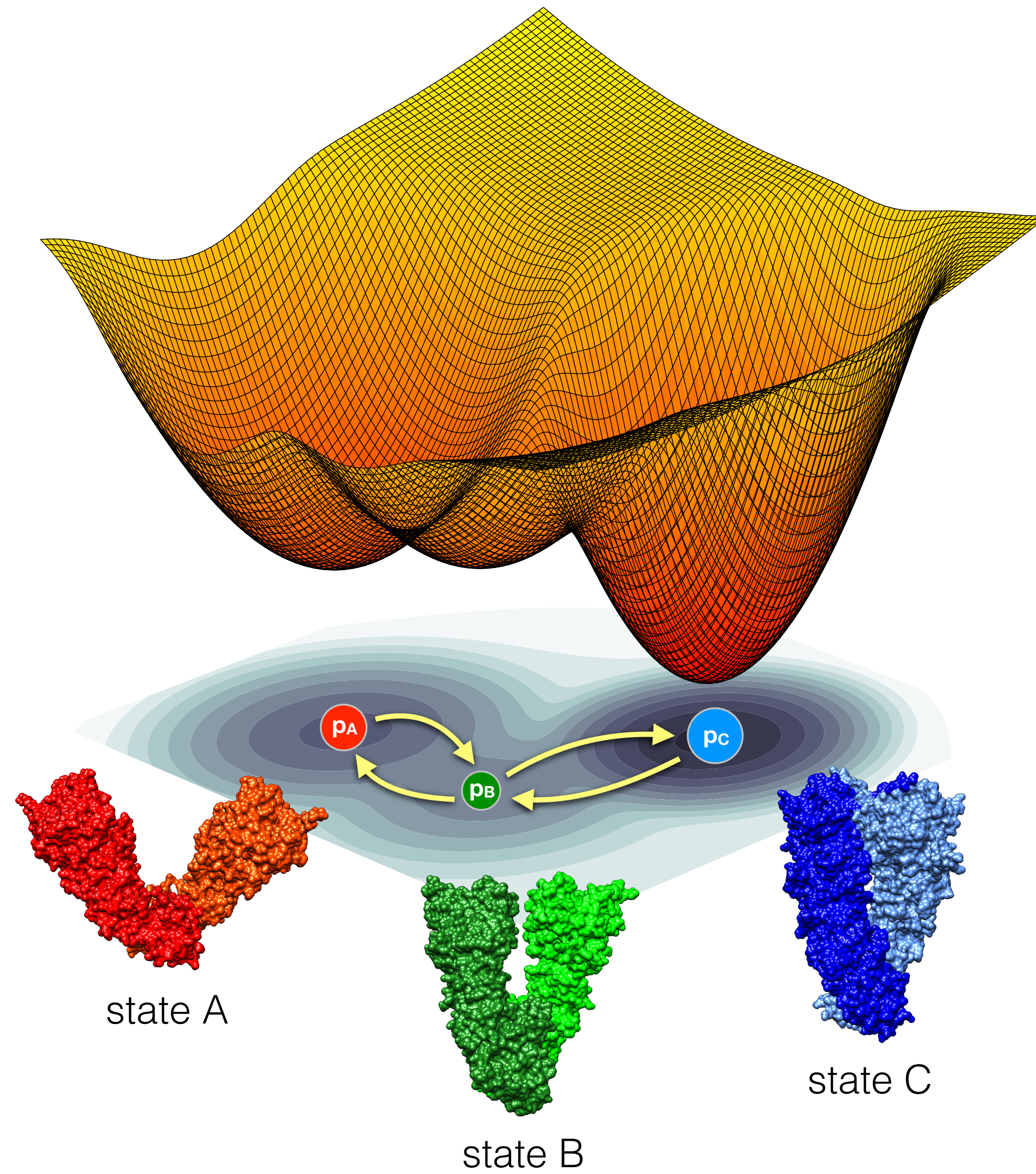


A Bayesian approach to integrate cryo-electron microscopy data into MD simulations with PLUMED

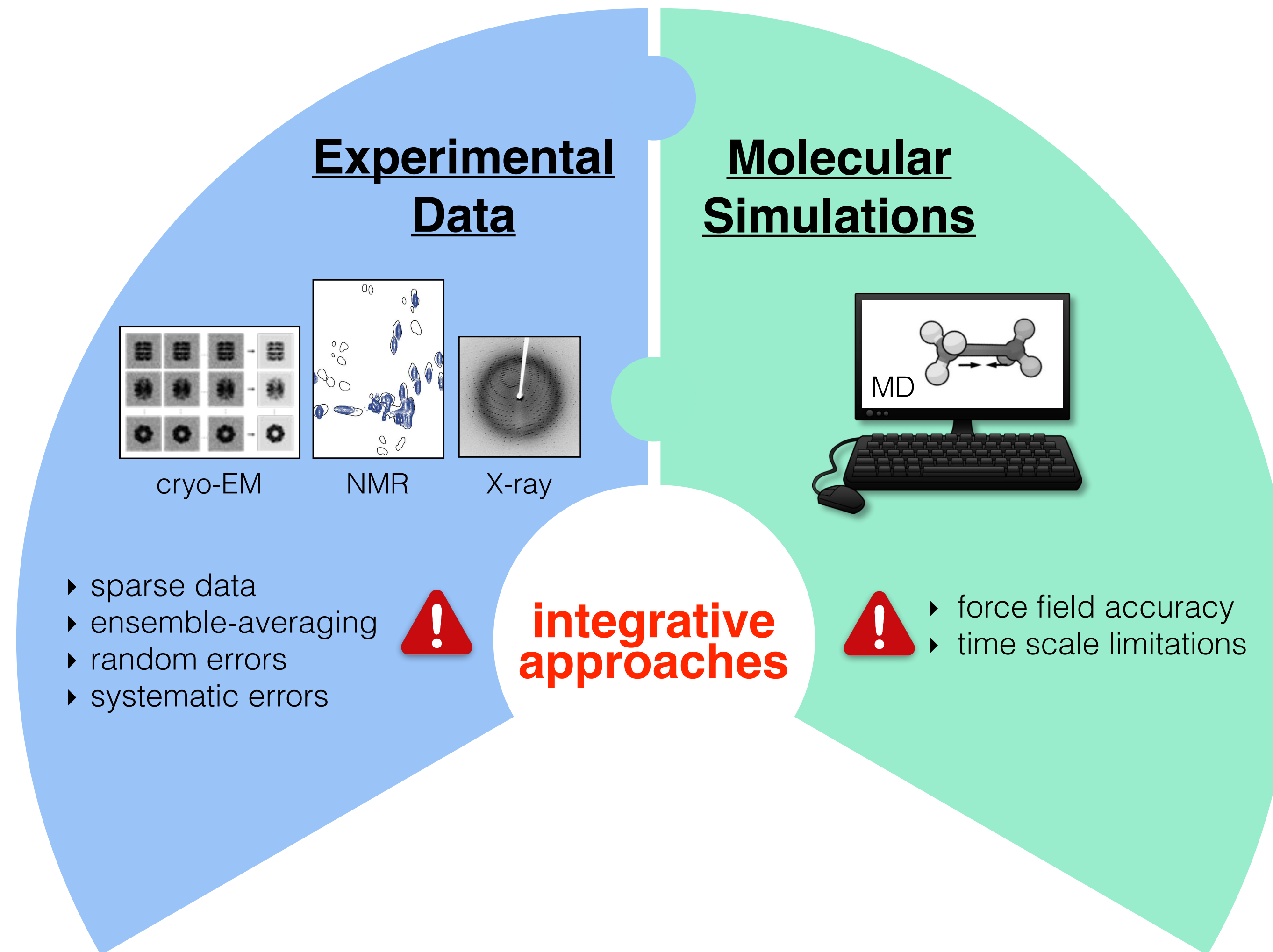
Max Bonomi
Institut Pasteur - CNRS
mbonomi@pasteur.fr

 @BonomiMax

Biological functions from structure and dynamics

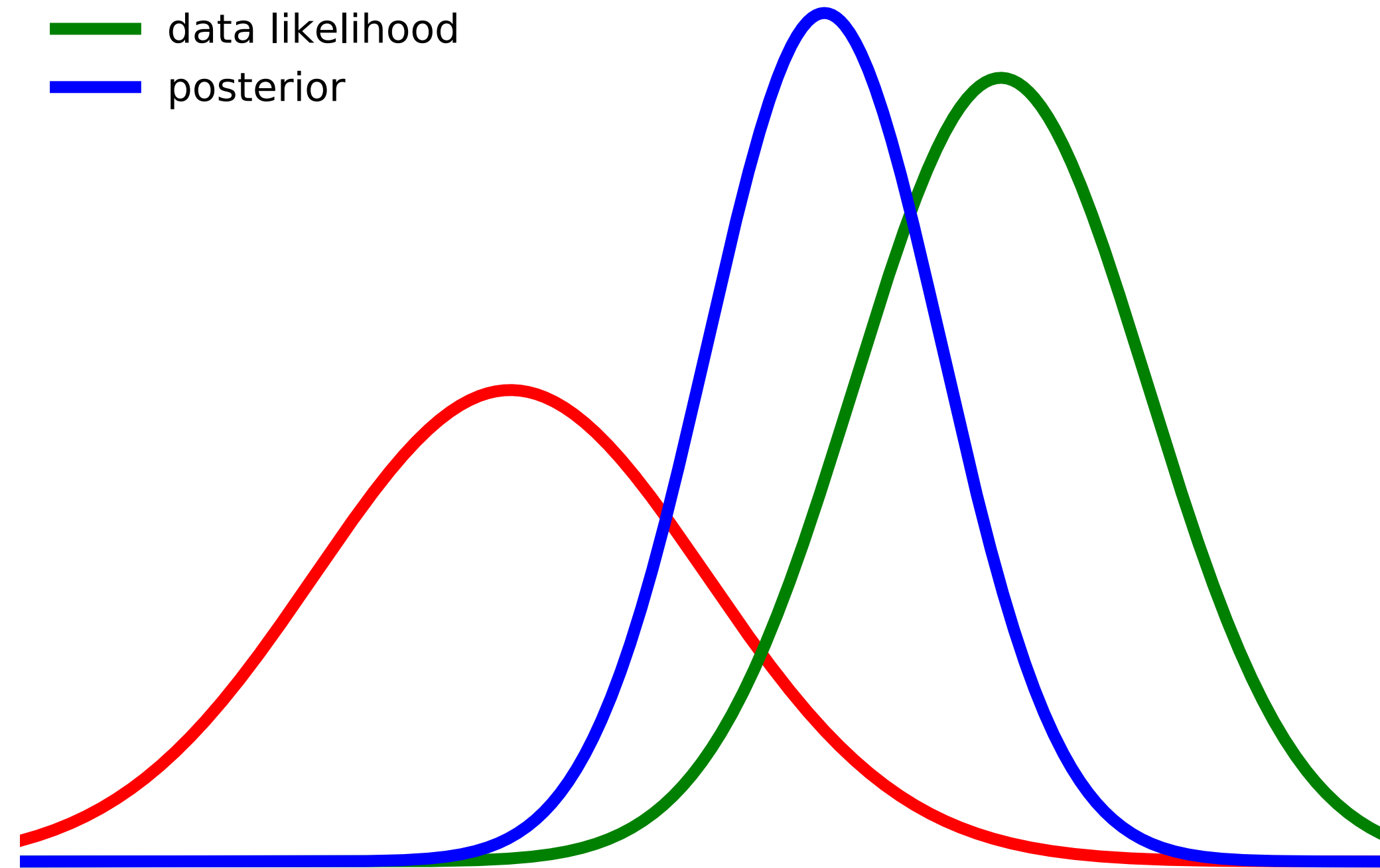


The best of both worlds: integrative or *hybrid* approaches



How? A little help from Bayes

— prior
— data likelihood
— posterior



Thomas Bayes (maybe)
1702-1761

$$p(X, \sigma | d) \propto p(d | X, \sigma) \cdot p(X) \cdot p(\sigma)$$

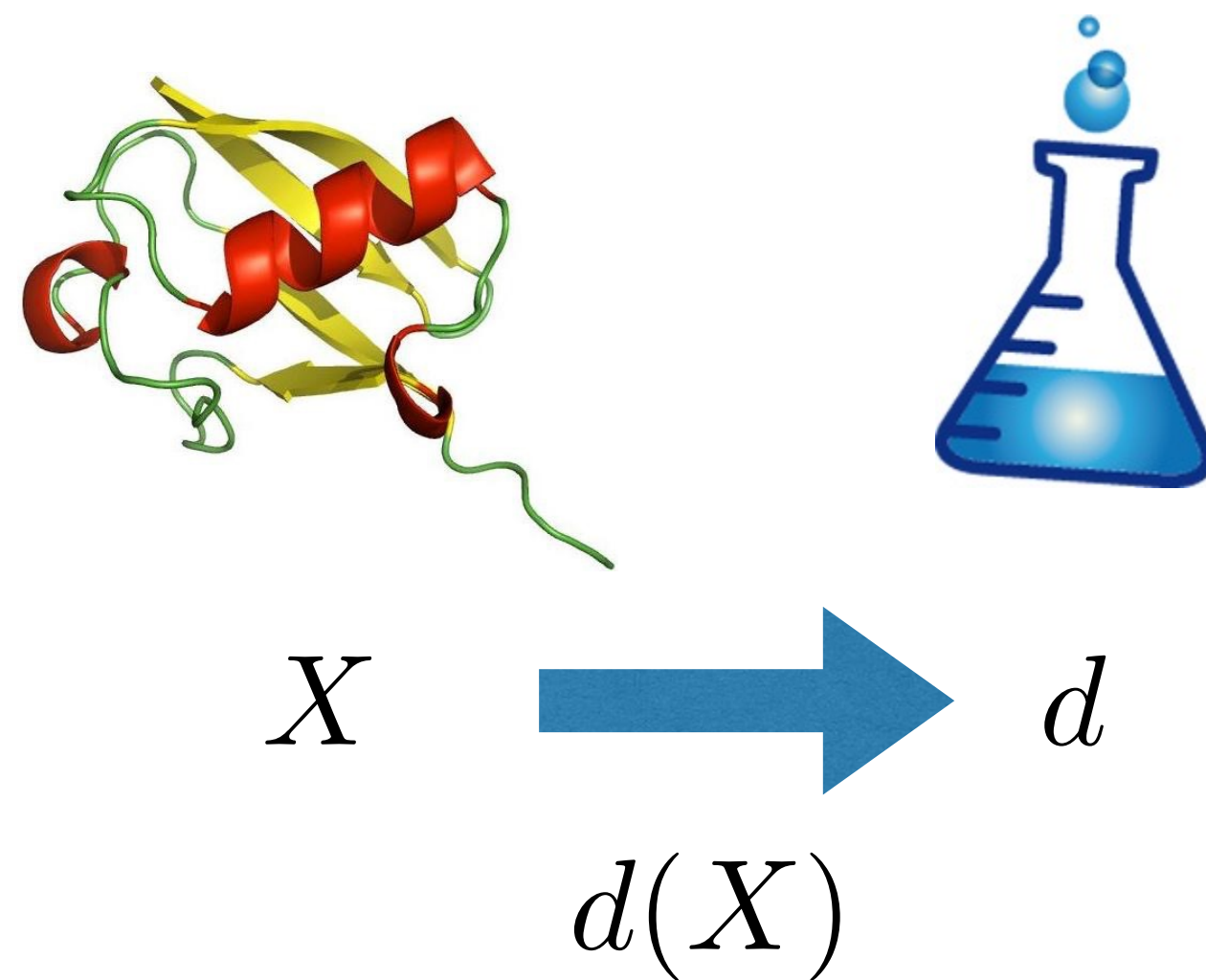
posterior

likelihood

priors

The ingredients I

1) Define a *forward model*, i.e. a predictor of the experimental data from a structural model:



Examples

NOE: $1/\text{distance}^6$

Chemical Shifts: SPARTA, SHIFTX, ...

SAXS: CRY SOL, FoXS, ...

Cross-links: distances, ...

Cryo-EM: Gaussians centered on atoms positions

2) Define a *noise model*, i.e. a model of the deviation between predicted and measured data. Typically Gaussian or LogNormal distribution:

$$p(d|X, \sigma, I) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{[d - d(X)]^2}{2\sigma^2}}$$

large error = high tolerance for deviations

small error = small tolerance for deviations



errors in experimental
data **and** predictors

The ingredients II

3) Choose the *priors*, i.e. your favorite MD force field

$$p(X|I) = e^{-\frac{E_{FF}(X)}{k_B T}}$$

and a prior for the uncertainty parameter(s):

$$p(\sigma|I) = \frac{1}{\sigma}$$

Jeffrey's prior

$$p(\sigma_i|\sigma_0, I) = \frac{2\sigma_0}{\sqrt{\pi}\sigma_i^2} \exp\left(-\frac{\sigma_0^2}{\sigma_i^2}\right)$$

Outliers prior

4) Sample (MD, MC, or combined methods) using the following potential
hybrid energy function:

large error = weak restraint

small error = strong restraint

$$E = -k_B T \log p(X, \sigma|d) = E_{FF}(X) + \frac{k_B T}{2\sigma^2} [d - d(X)]^2 + E(\sigma)$$

hybrid
energy

MD
force field

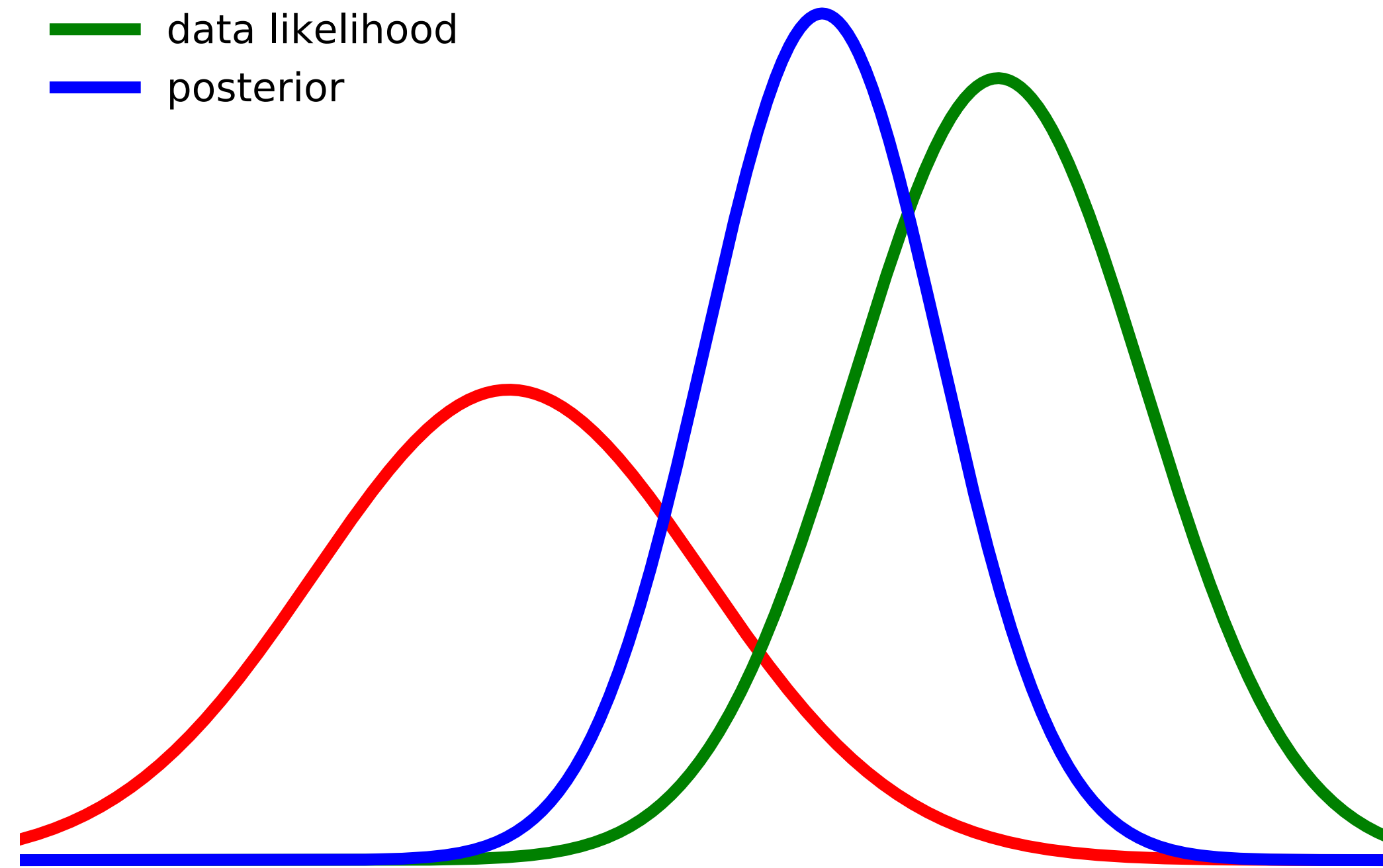
data
restraint

error
restraint

Inferential structure determination

Rieping et al. Science 2005

— prior
— data likelihood
— posterior



$$p(X, \sigma | d) \propto \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{[d - d(X)]^2}{2\sigma^2}} \cdot e^{-\frac{E_{FF}(X)}{k_B T}} \cdot p(\sigma)$$

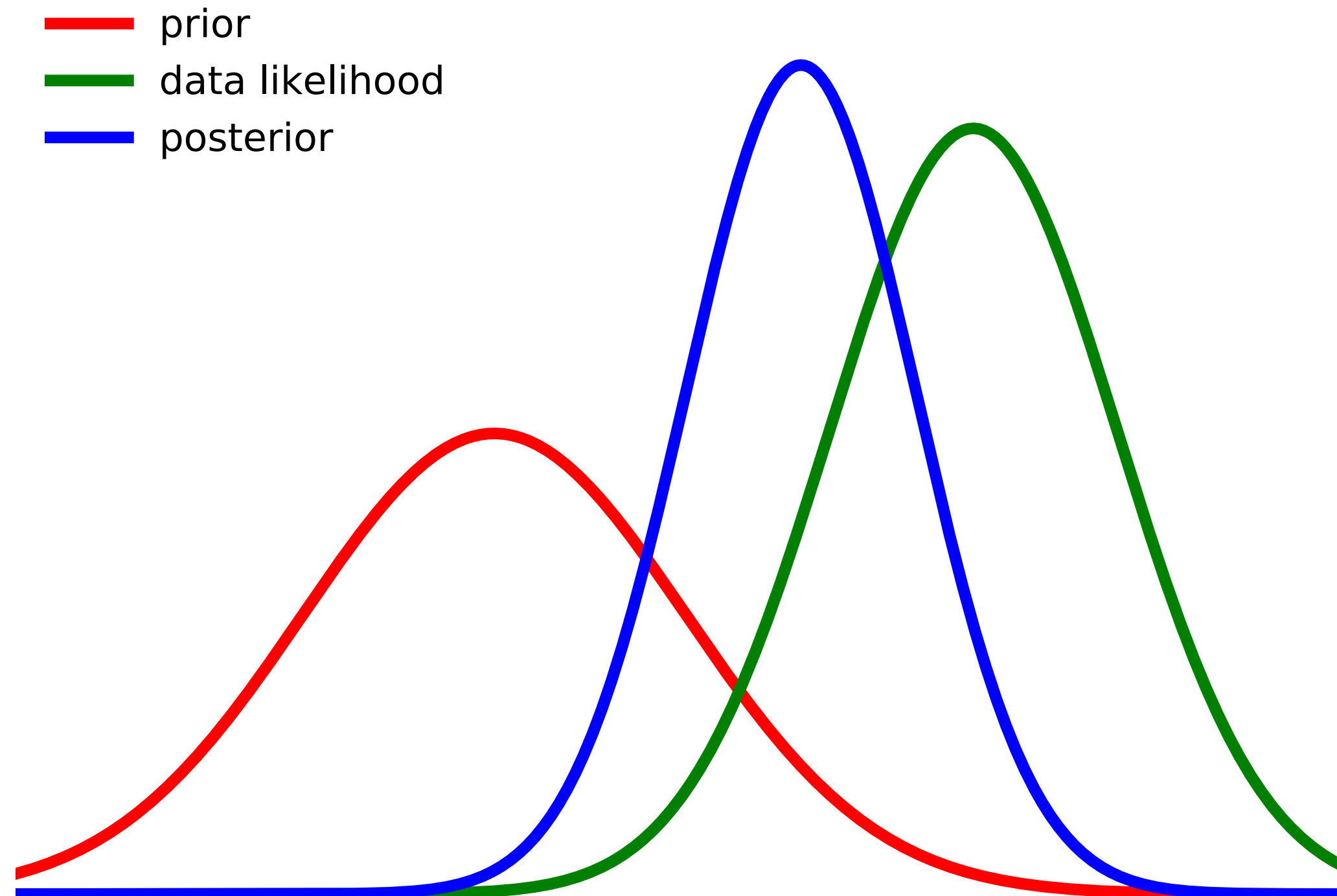
posterior

likelihood

priors

Inferential structure determination

Rieping *et al.* Science 2005



$$E = -k_B T \log p(X, \sigma | d) = E_{FF}(X) + \frac{k_B T}{2\sigma^2} [d - d(X)]^2 + E(\sigma)$$

hybrid
energy

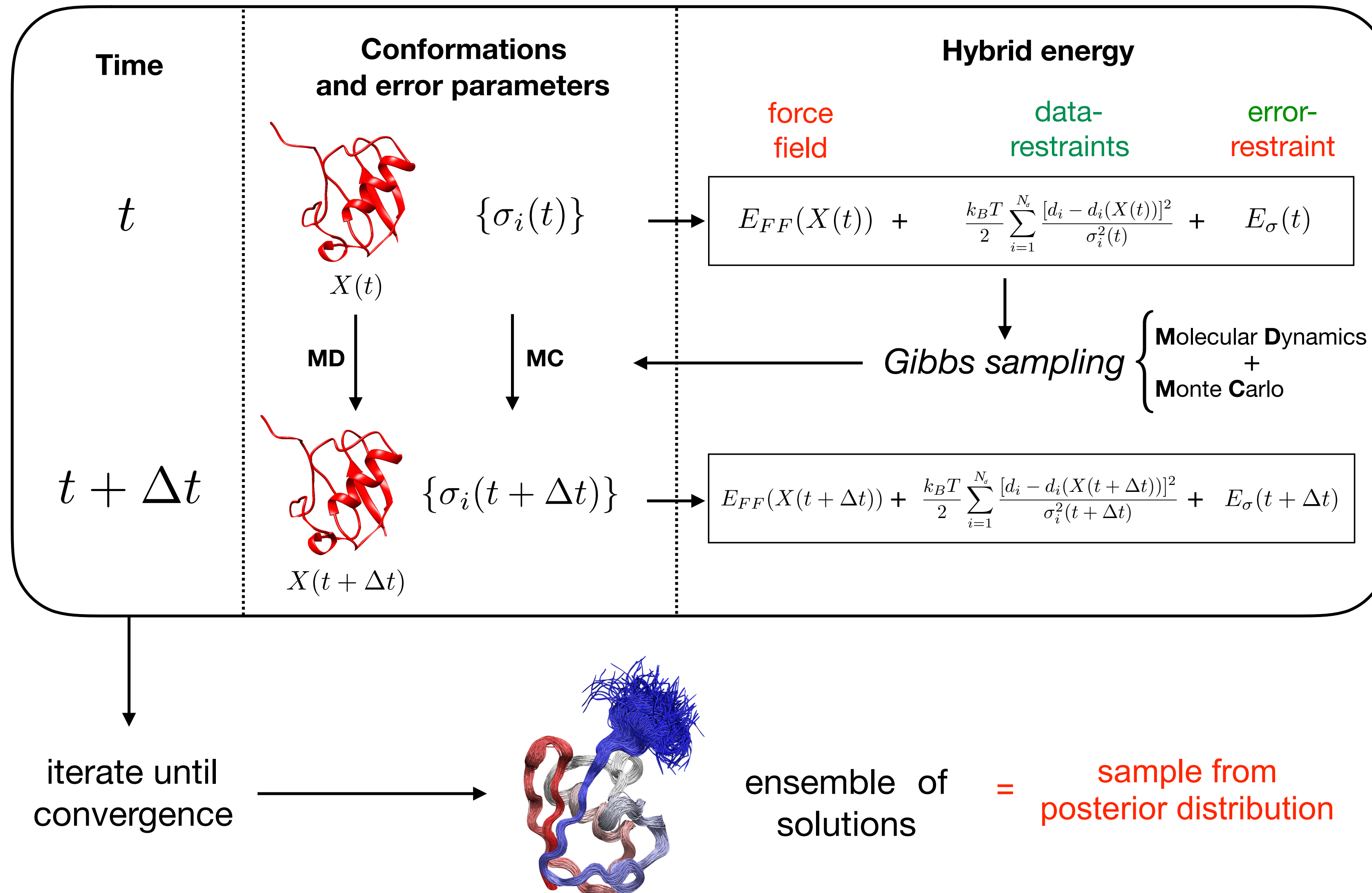
MD
force field

data
restraint

error
restraint

Inferential structure determination

Rieping *et al.* Science 2005



The advantages of the Bayesian approach

- Information other than experimental data can be rigorously incorporated into the modelling
- Errors in the data are accounted for

An *uncertainty parameter* σ is associated to the data and inferred along with the structure X

$$p(X, \sigma | D, I) \propto p(D | X, \sigma, I) \cdot p(\sigma | I) \cdot p(X | I)$$

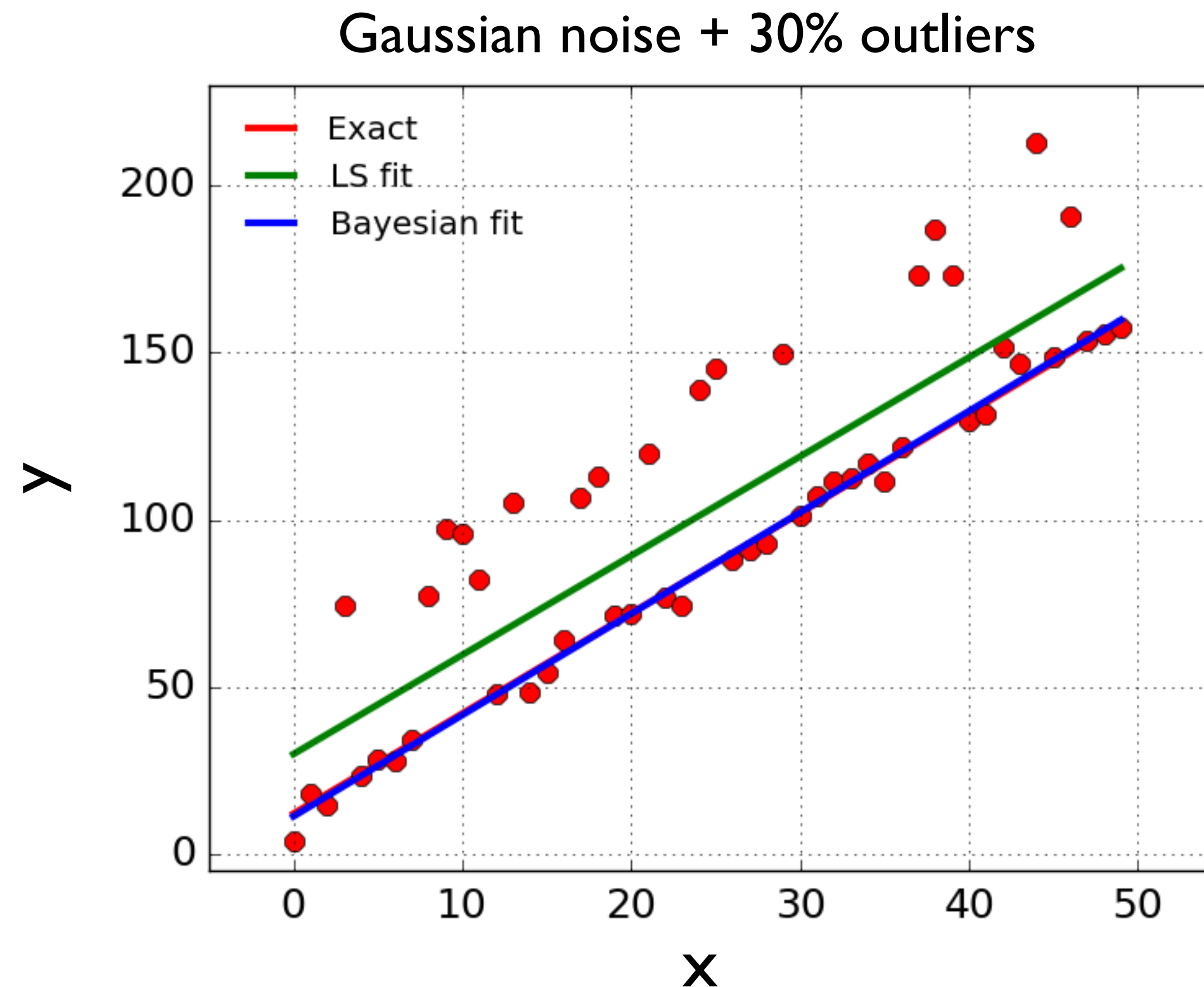
- Data from different experiments is automatically balanced

$$p(X, \{\sigma_i\} | \{d_i\}, I) \propto \left[\prod_i p(d_i | X, \sigma_i, I) \cdot p(\sigma_i | I) \right] \cdot p(X | I)$$

The *weight* of each experimental data point in the construction of the structural model(s) is automatically determined based on its accuracy

Example: Bayesian linear fit

Let's suppose our data can be described by a linear relation, with some error



$$y = Ax + B$$

$$A_0 = 3.0 \quad B_0 = 12.0$$

$$A_{LS} = 2.9 \quad B_{LS} = 29.8$$

$$A_B = 3.0 \quad B_B = 11.3$$

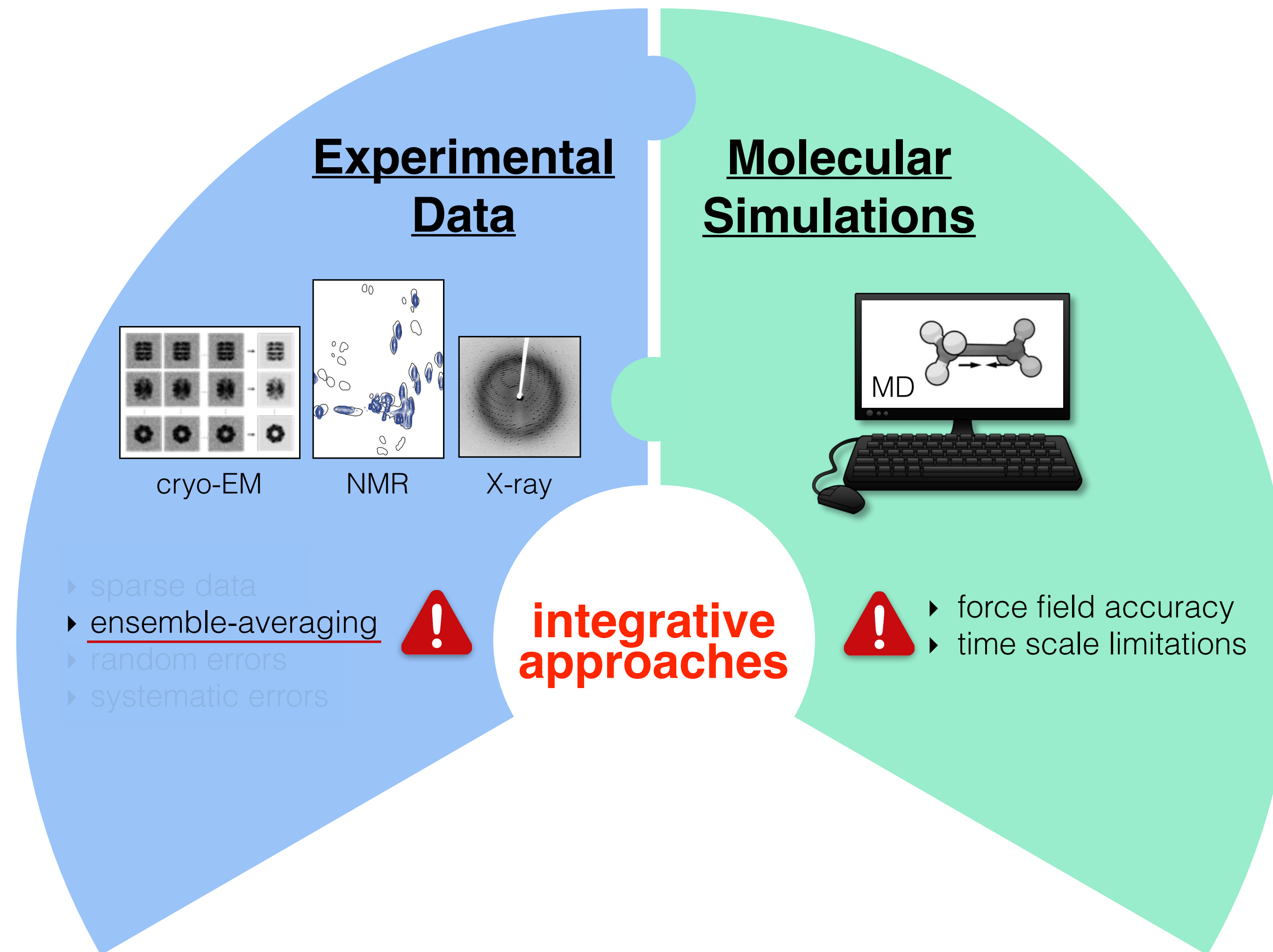
$$p(A, B, \sigma_0 | \{Y_k\}, I) \propto \prod_k \frac{1}{\sigma_k \sqrt{2\pi}} e^{-\frac{[f(x_k) - Y_k]^2}{2\sigma_k^2}} \cdot \frac{2\sigma_0}{\sqrt{\pi}\sigma_k^2} e^{-\frac{\sigma_0^2}{\sigma_k^2}}$$

Posterior probability

$$f(x_k) = Ax_k + B$$

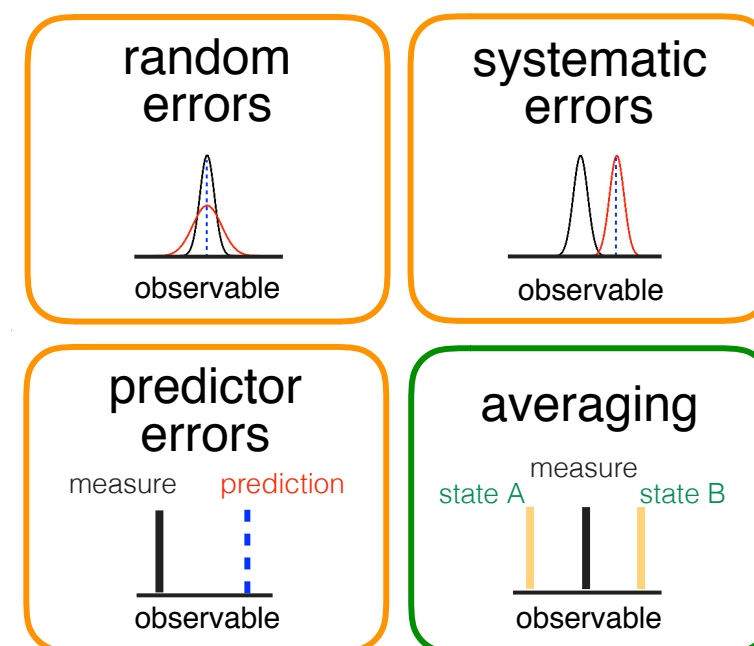
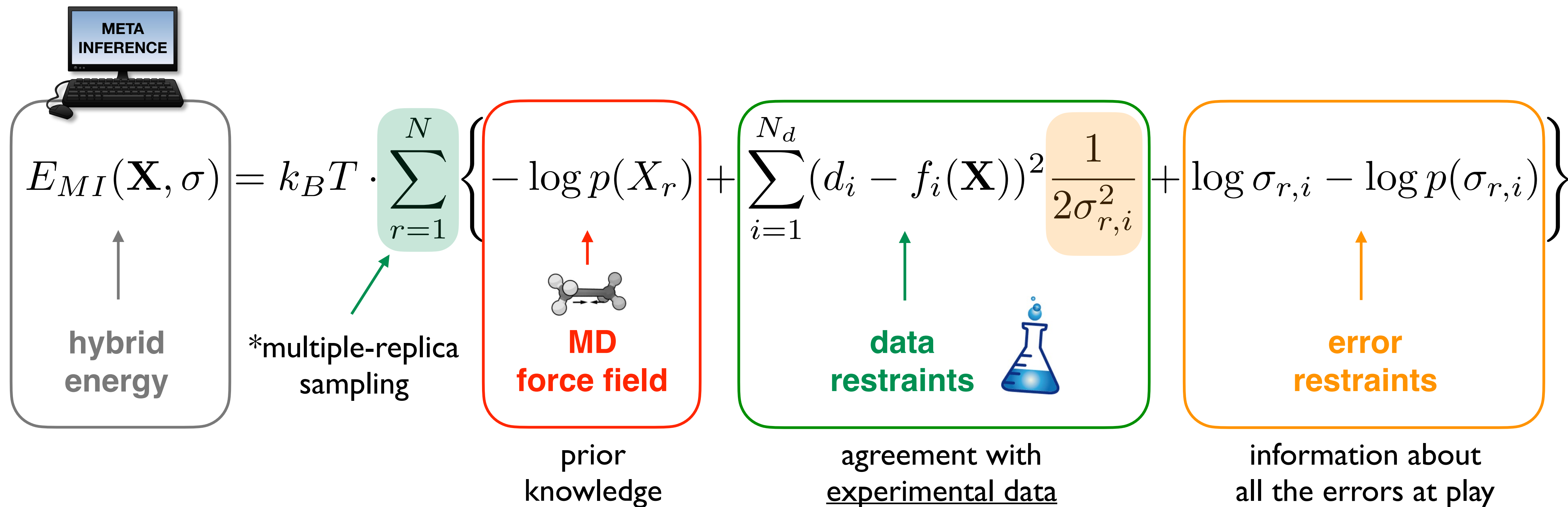
Linear forward model

The best of both worlds: integrative or *hybrid* approaches



Inferential determination of structural ensembles with MetaInference

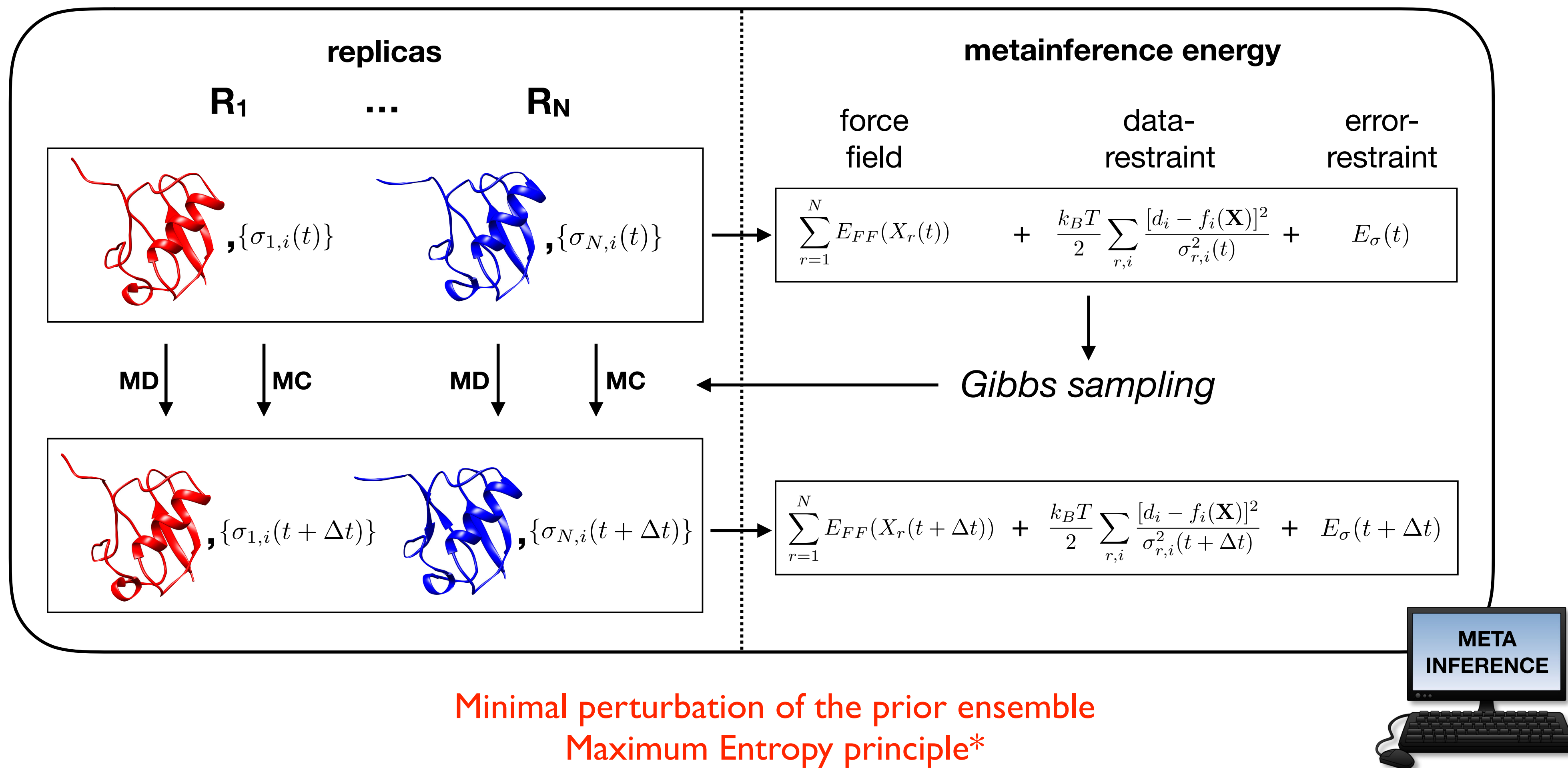
Bonomi *et al.* Sci Adv 2016



*Lindorff-Larsen *et al.* Nature 2005
 Cavalli, Camilloni, Vendruscolo. J Chem Phys 2013
 Hummer, Köfinger. J Chem Phys 2015

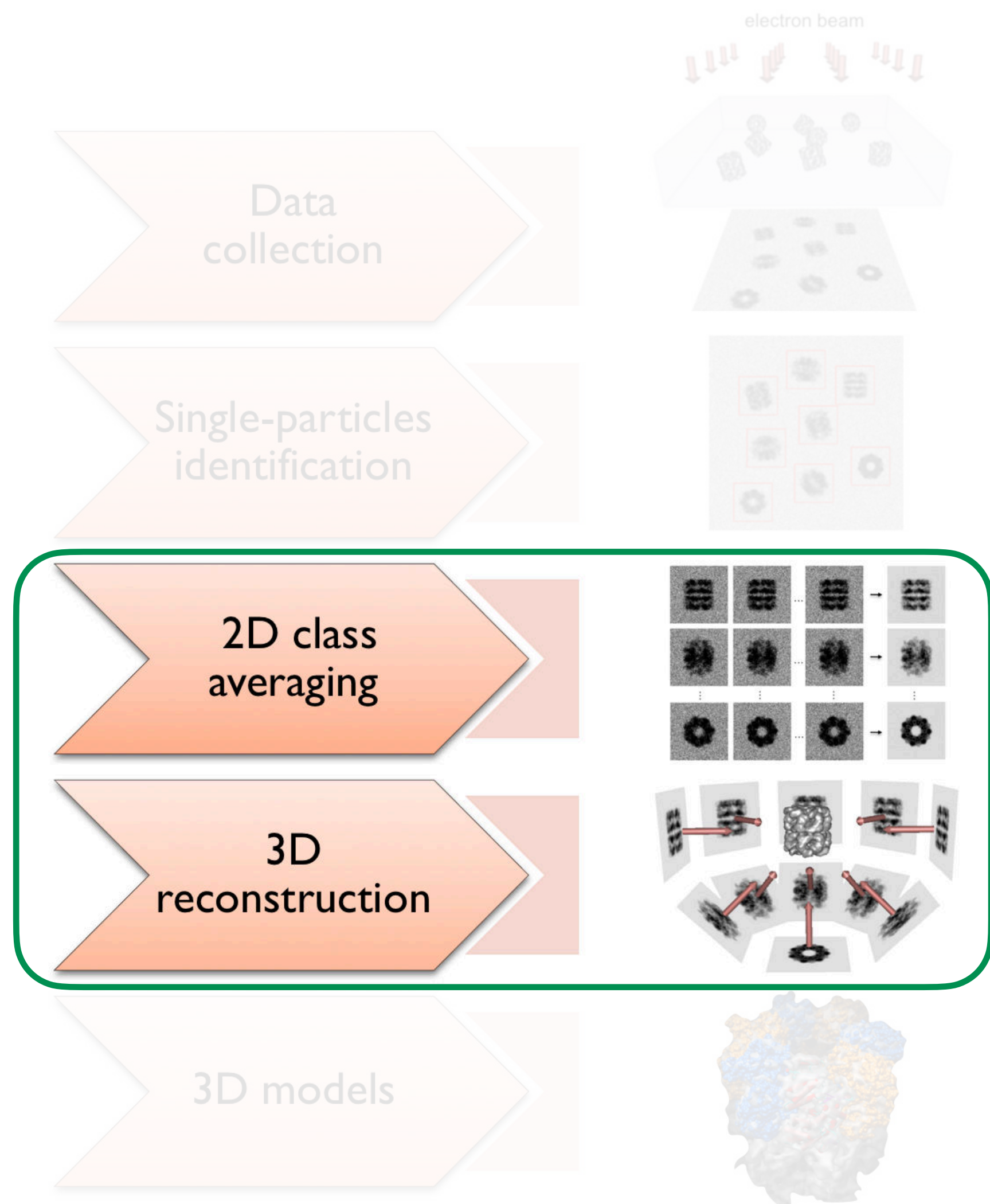
Inferential determination of structural ensembles with Metainference

Bonomi *et al.* Sci Adv 2016

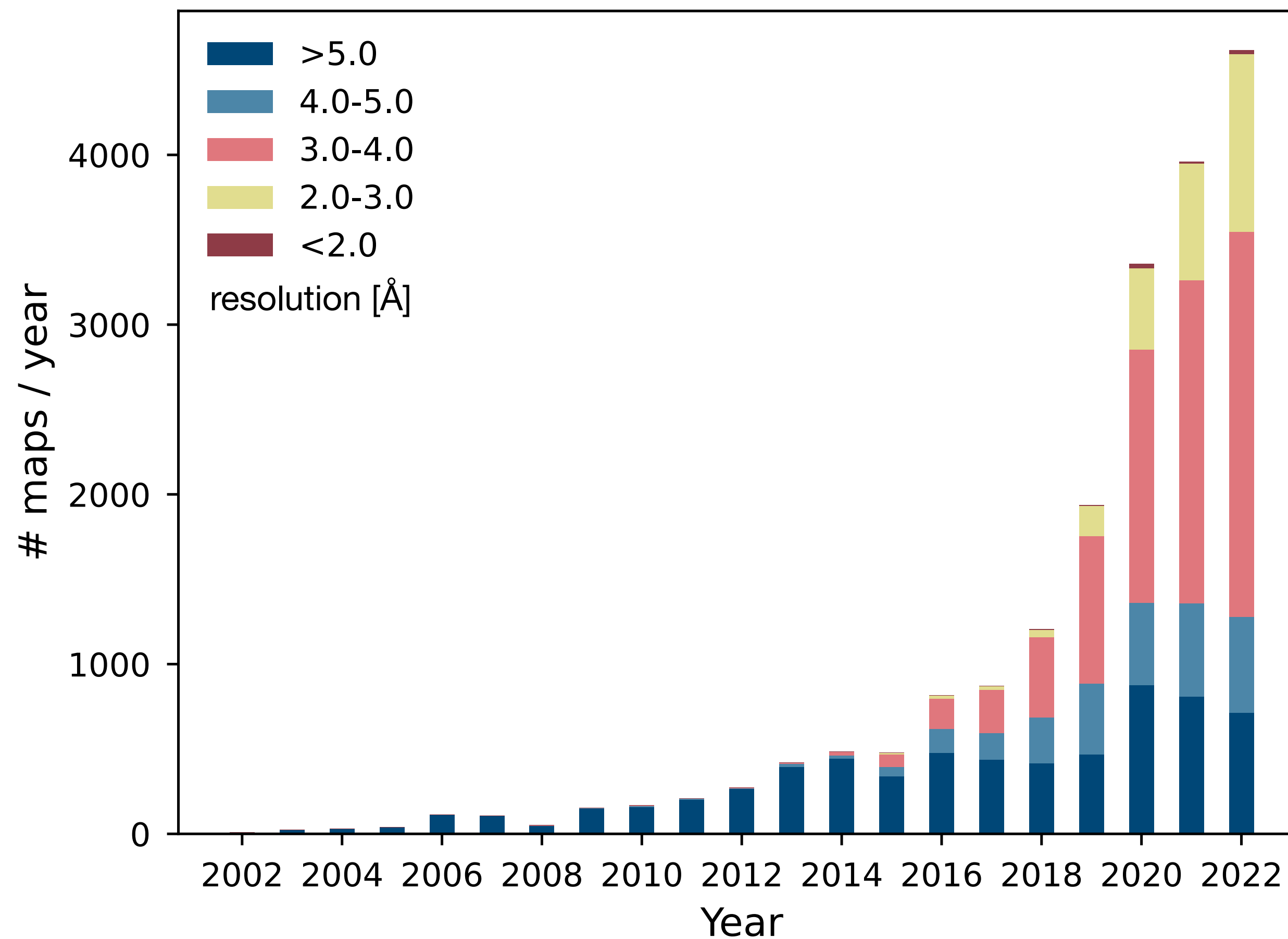


*see Hummer & Köfinger. J Chem Phys 2015

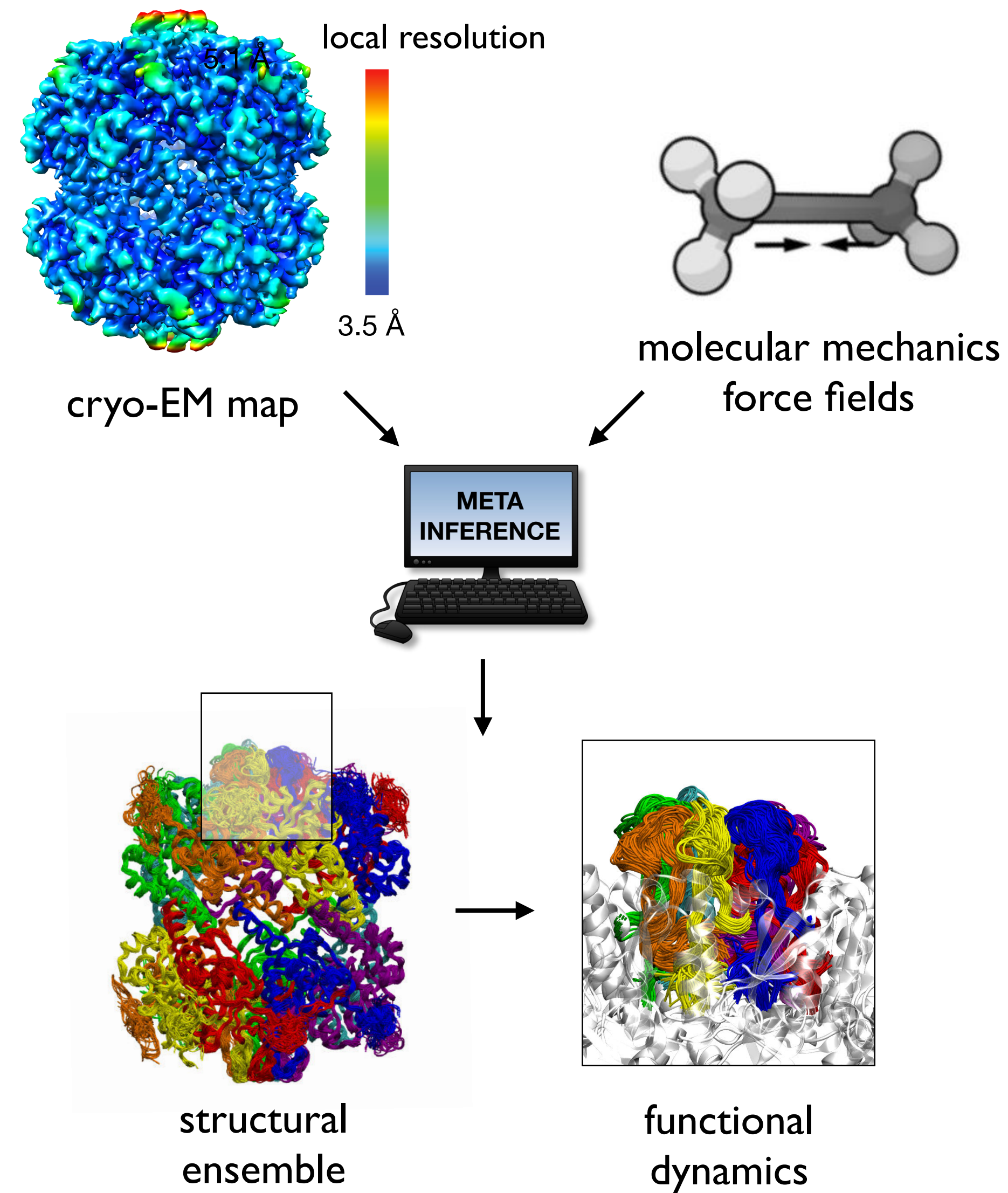
The cryo-EM resolution revolution



Number of single-particle maps released between 2002 and 2022



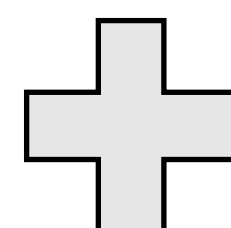
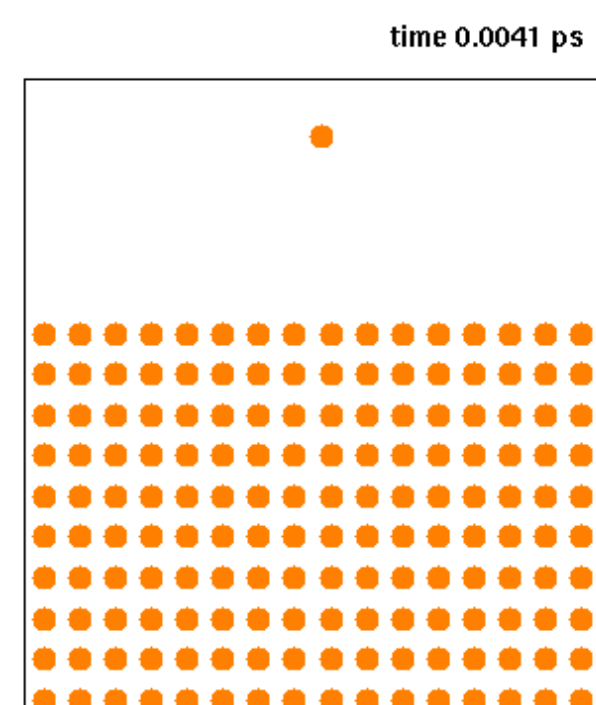
The cryo-EM resolution revolution



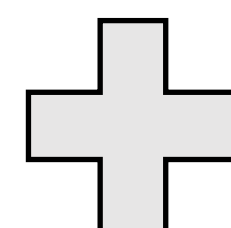
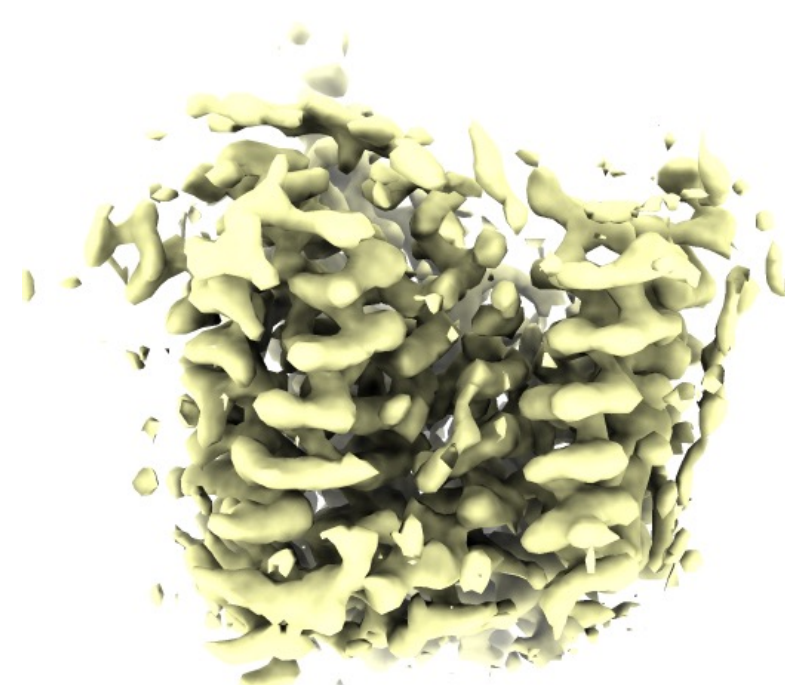
Integrate cryo-EM into MD simulations with EMMIVox

Hoff & Bonomi. In preparation; Bonomi, ..., Pellarin. Structure 2019

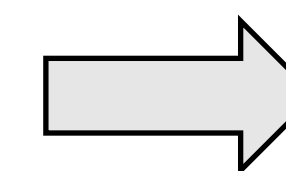
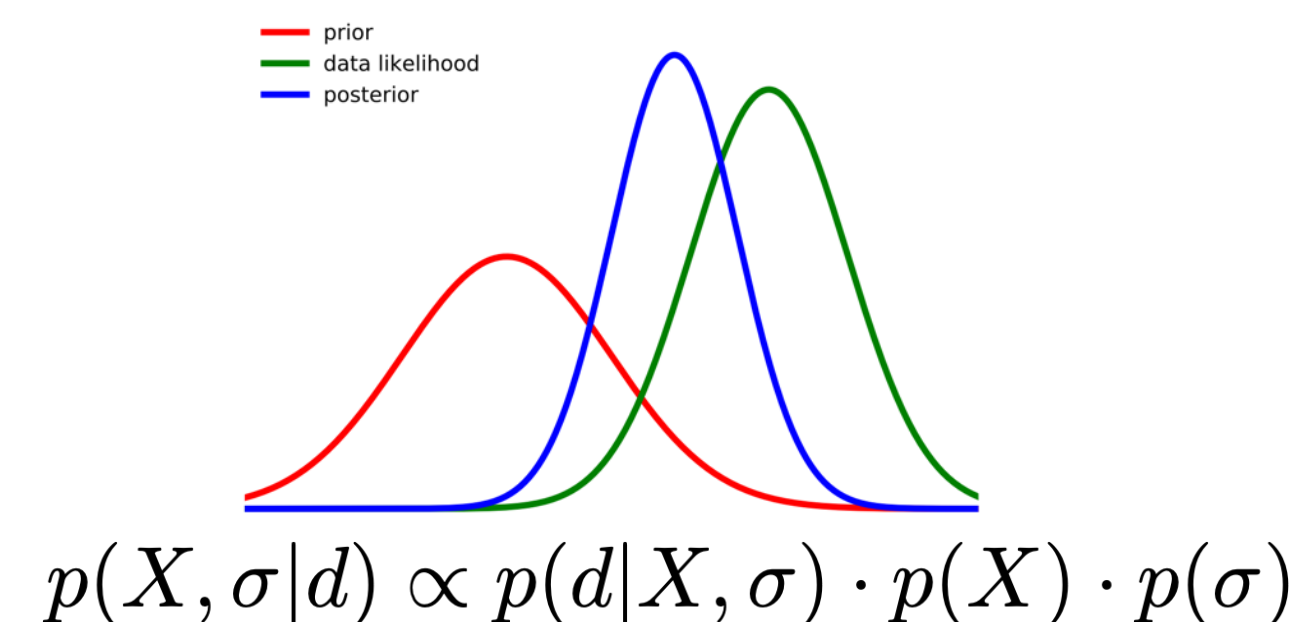
Accuracy of MD is limited!



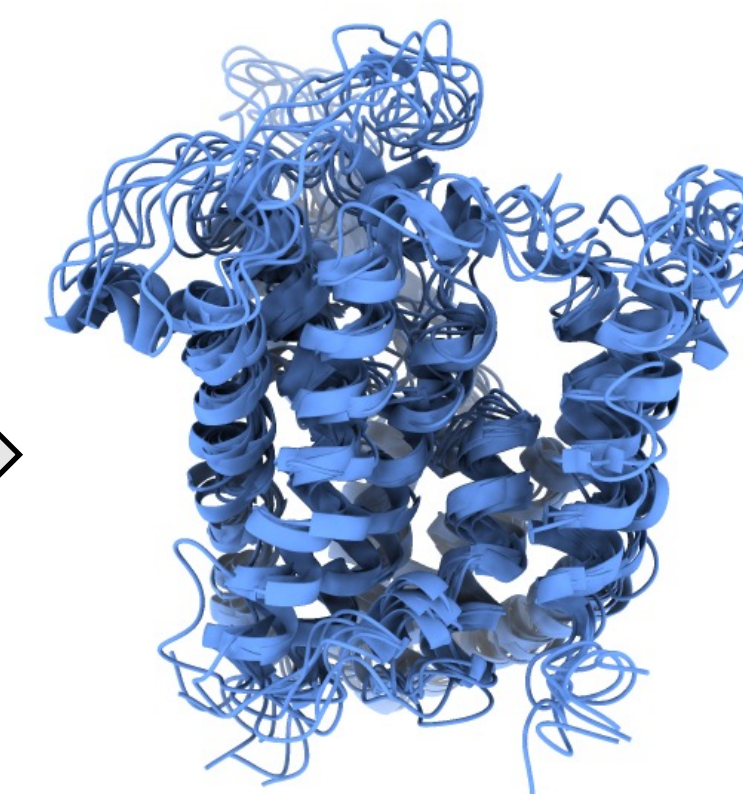
Experimental data



Bayesian inference



Structural models



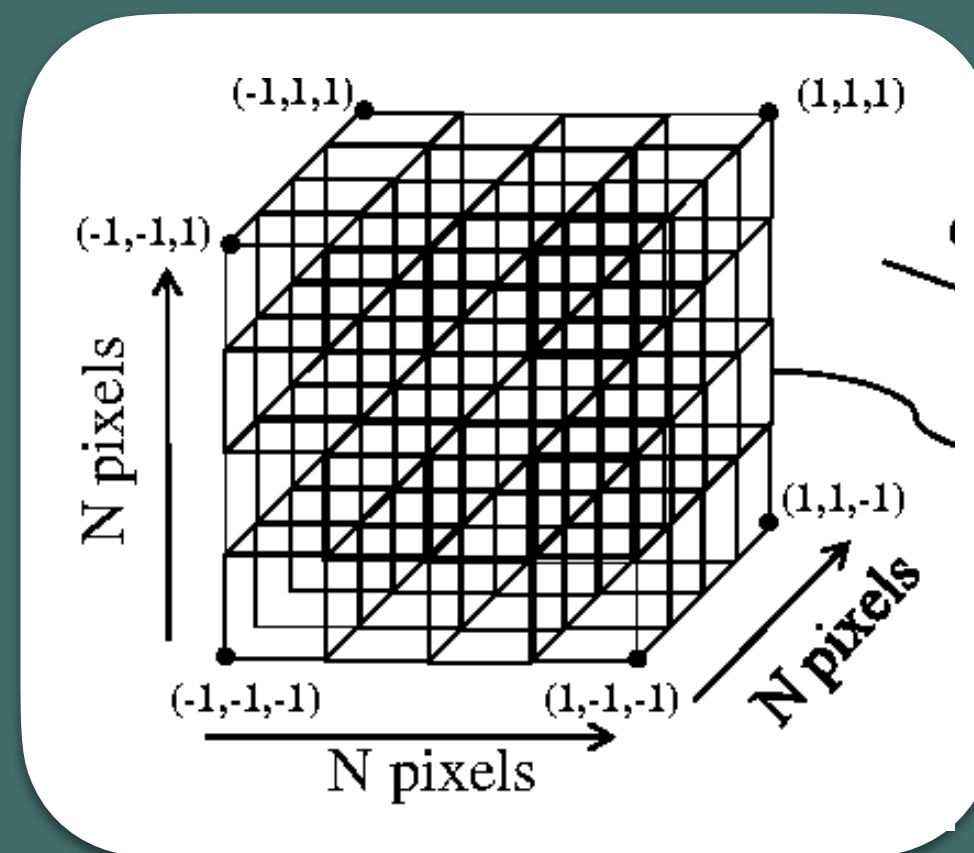
Objectives:

- Improving stereochemical quality of deposited PDBs (fewer clashes, ...)
- Modelling structural ensembles with MetaInference
- Integrating other experimental data (NMR, SAXS, EPR, ...)

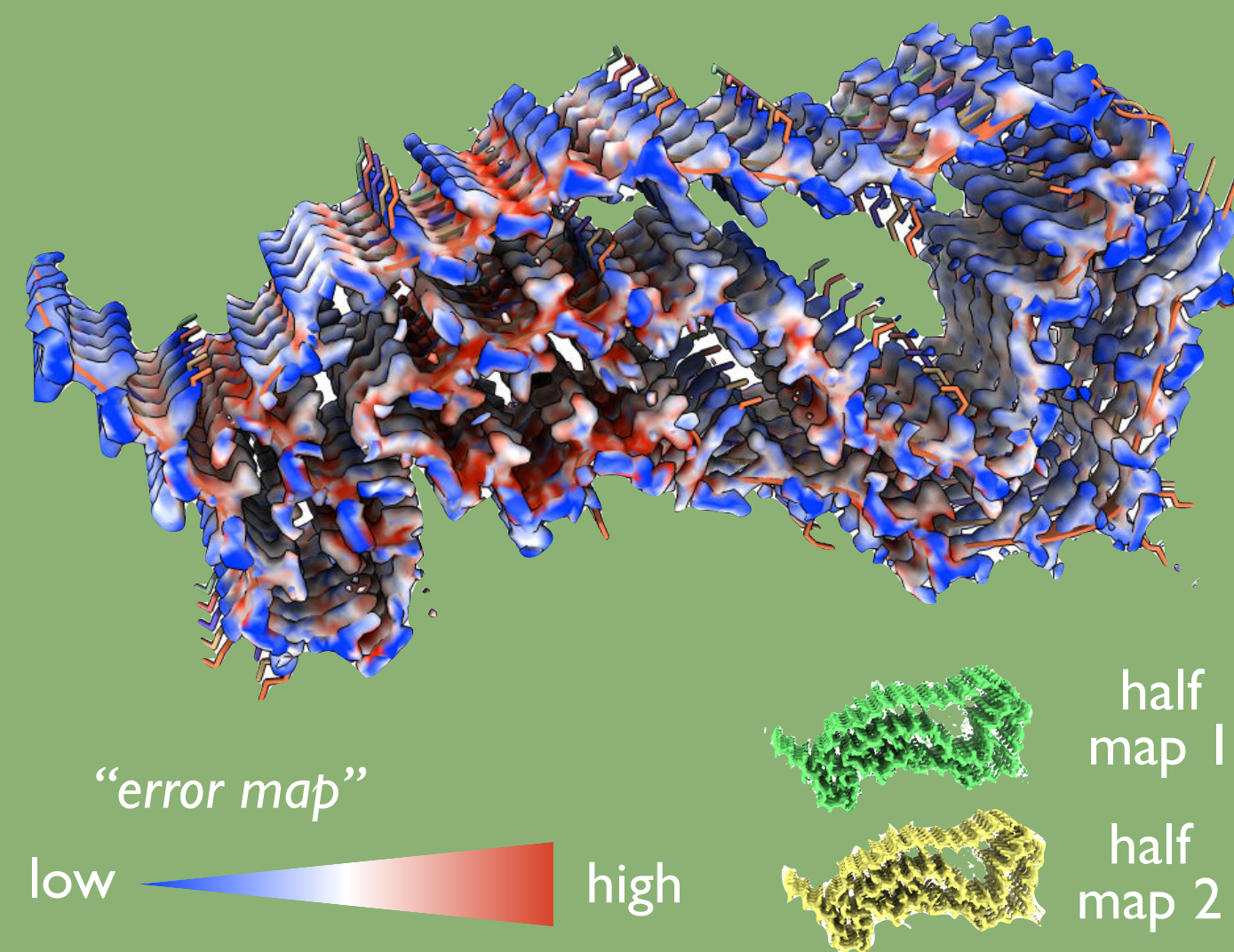
Integrate cryo-EM into MD simulations with EMMIVox

Hoff & Bonomi. In preparation; Bonomi, ..., Pellarin. Structure 2019

Estimate correlation
between voxels



Estimate
“experimental noise”
for each voxel



Gaussian noise model
for each voxel

$$p(\phi_{D,i}|M) = \frac{1}{\sqrt{2\pi}\sigma_i} \cdot \exp\left[-\frac{(\phi_{D,i} - \phi_{M,i})^2}{2\sigma_i^2}\right]$$

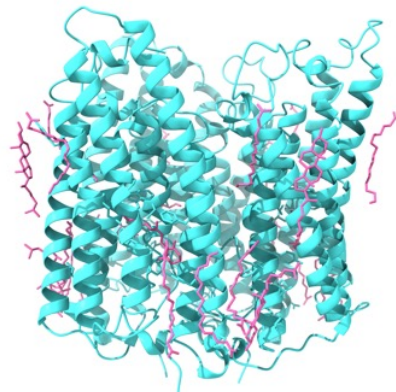
- remove correlated voxels
- one error parameter per voxel
- minimum value from error map
- error marginalization

Better balance between stereochemical quality and fit to the map

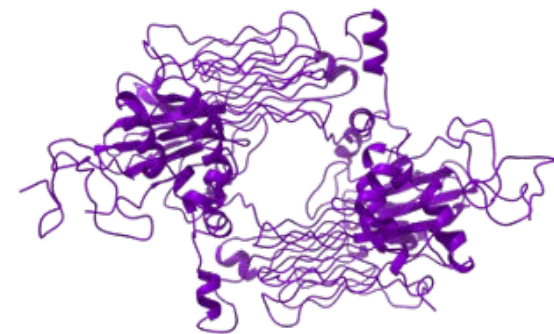
7p6a



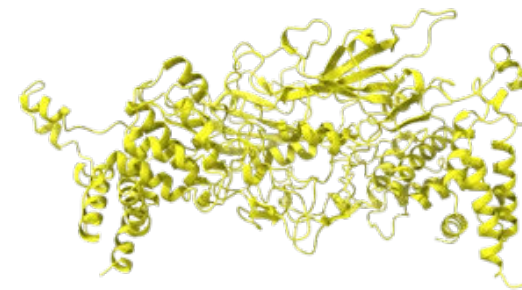
7w9w



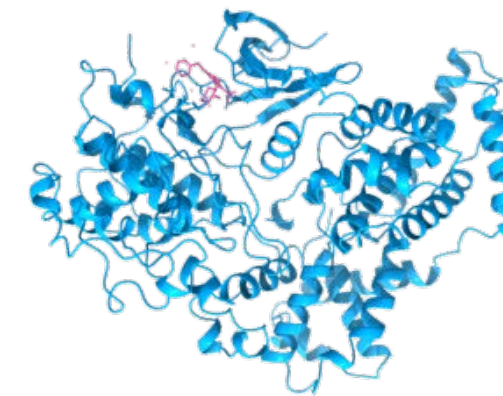
7n00



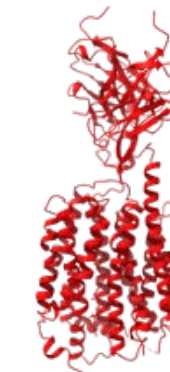
7t4n



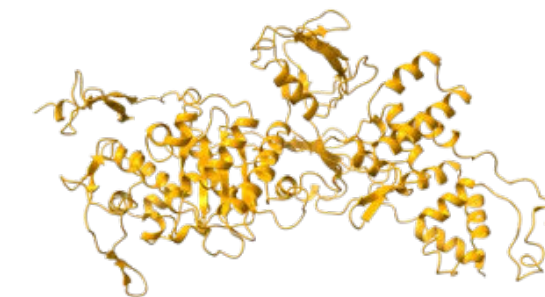
7b5o



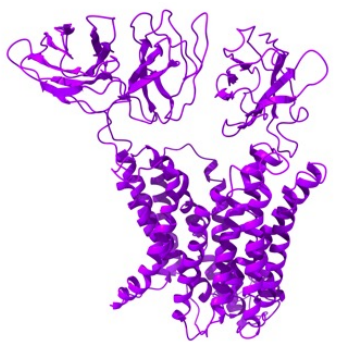
7mjs



7lq6



7nqk



Clash between atoms: Clashscore

Overall stereochemical quality: Molprobability

Fit model/map: CC_mask

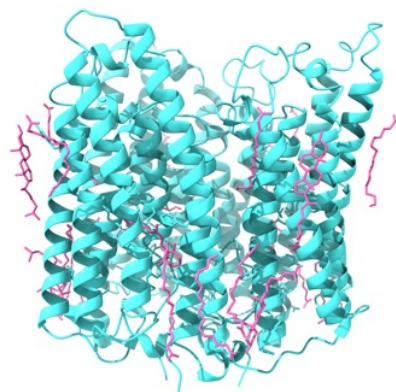
Correct positioning of sidechains: EMRinger

Better balance between stereochemical quality and fit to the map

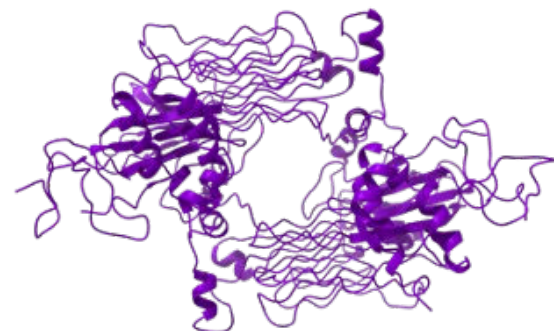
7p6a



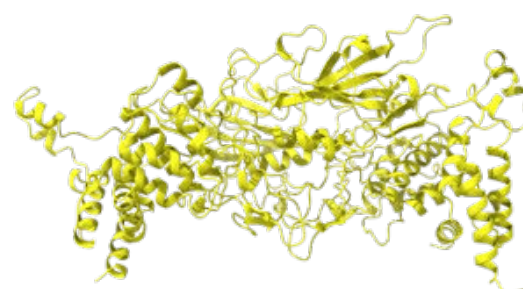
7w9w



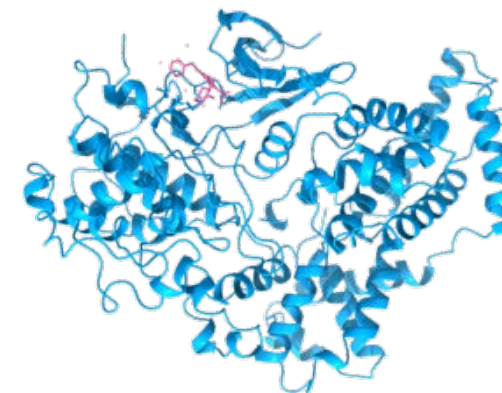
7n00



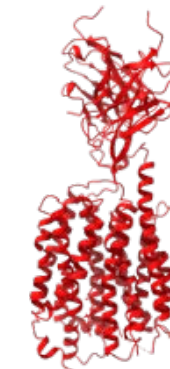
7t4n



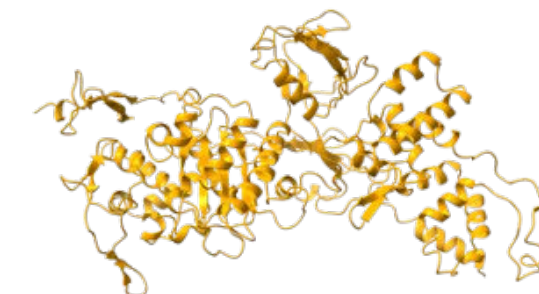
7b5o



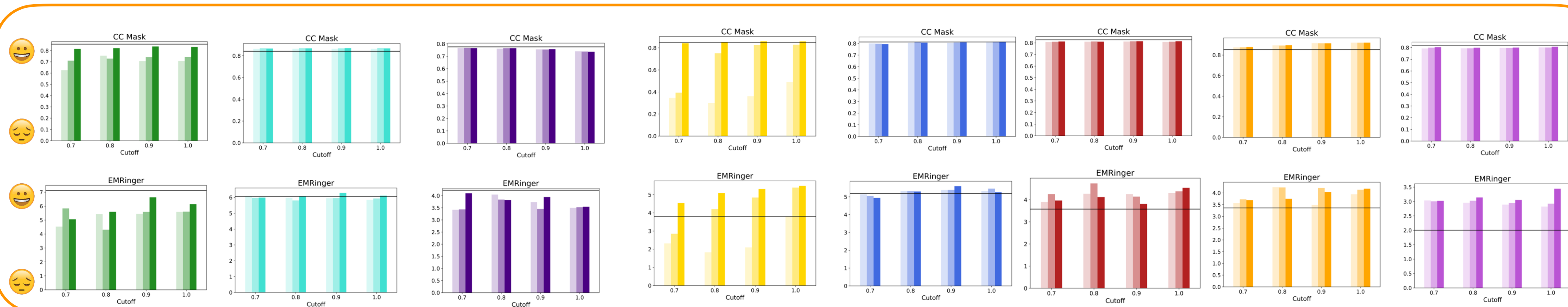
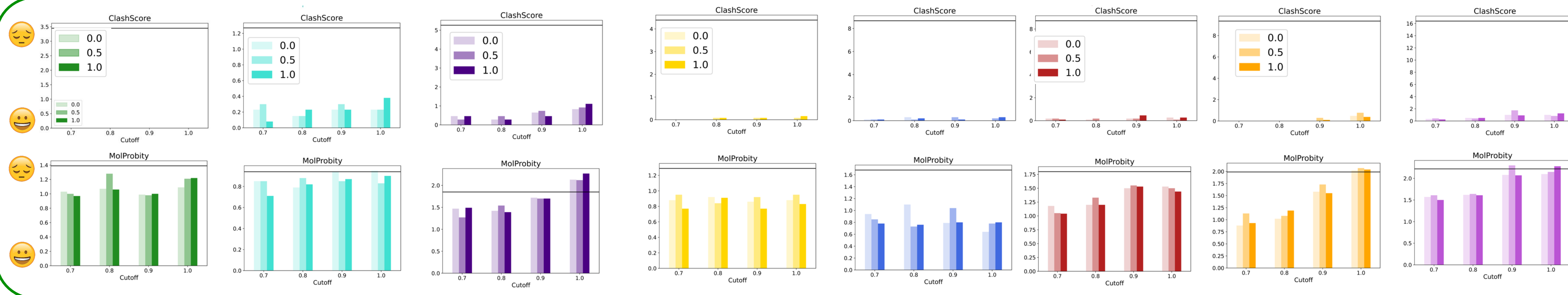
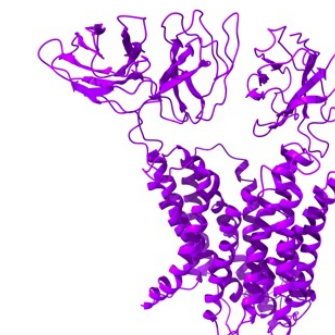
7mjs



7lq6



7nqk



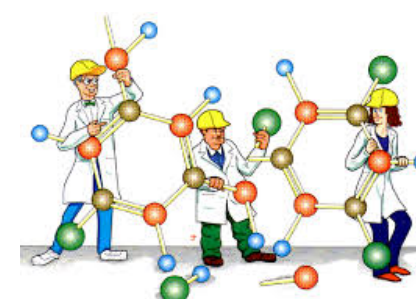
EMMIVox is implemented in PLUMED

Bonomi *et al.* CPC 2009; Tribello *et al.* CPC 2014

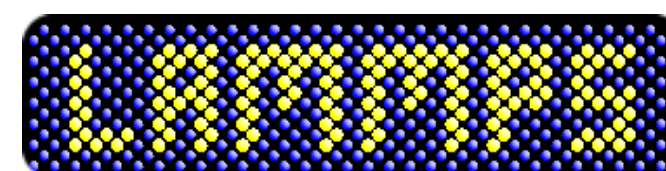
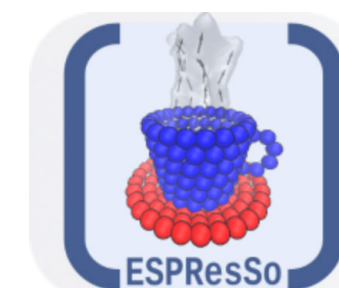


www.plumed.org

- open-source, freely-available
- community-developed
- enhanced sampling methods for MD
- analysis tools
- algorithms for integrative biology



DL_POLY

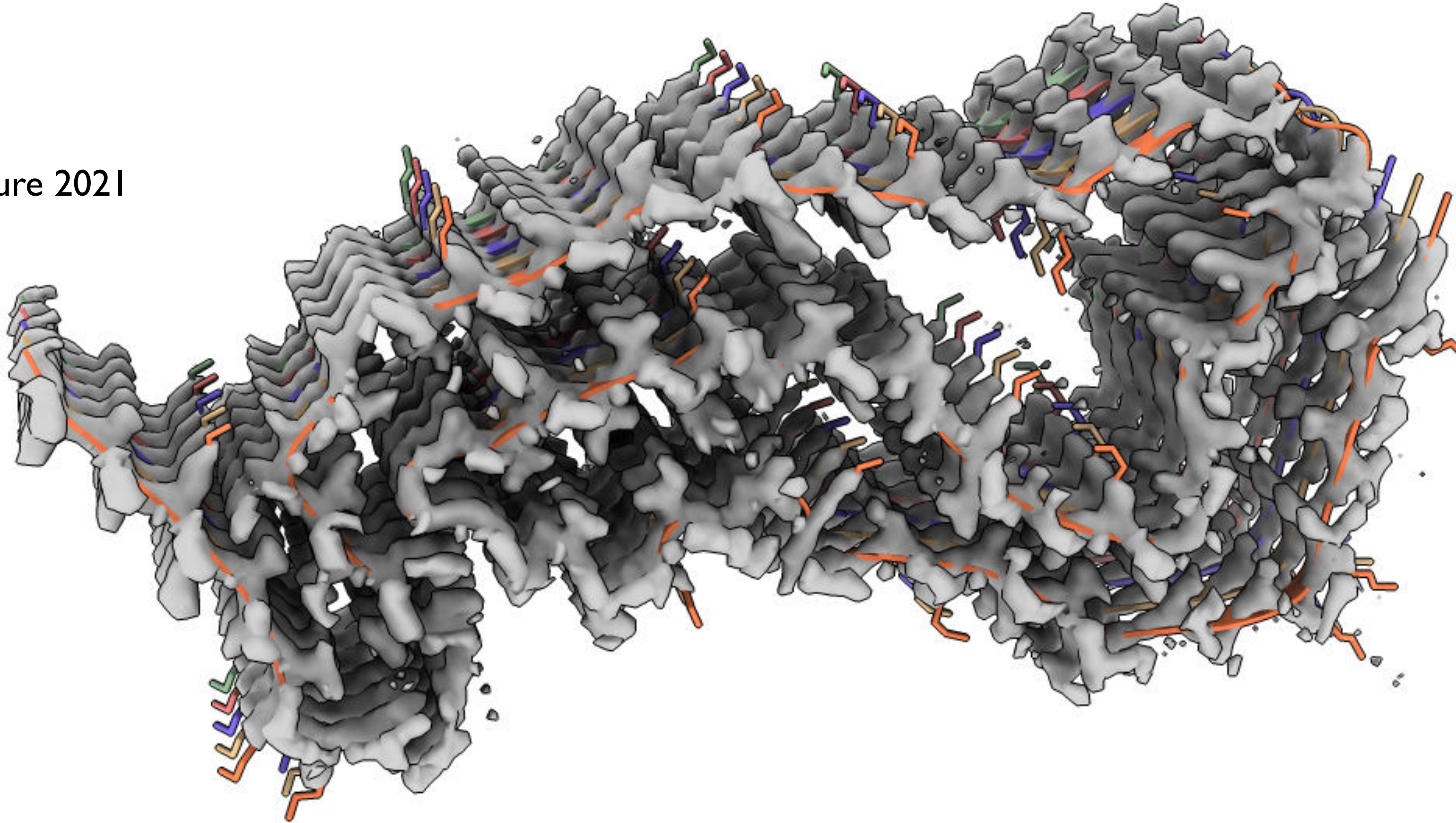


Today's system

Tau filament from the brains of individuals with PSP

*PDB 7p6a
EMDB I3223
resolution 1.9 Å

*Shi, ..., Scheres. Nature 2021

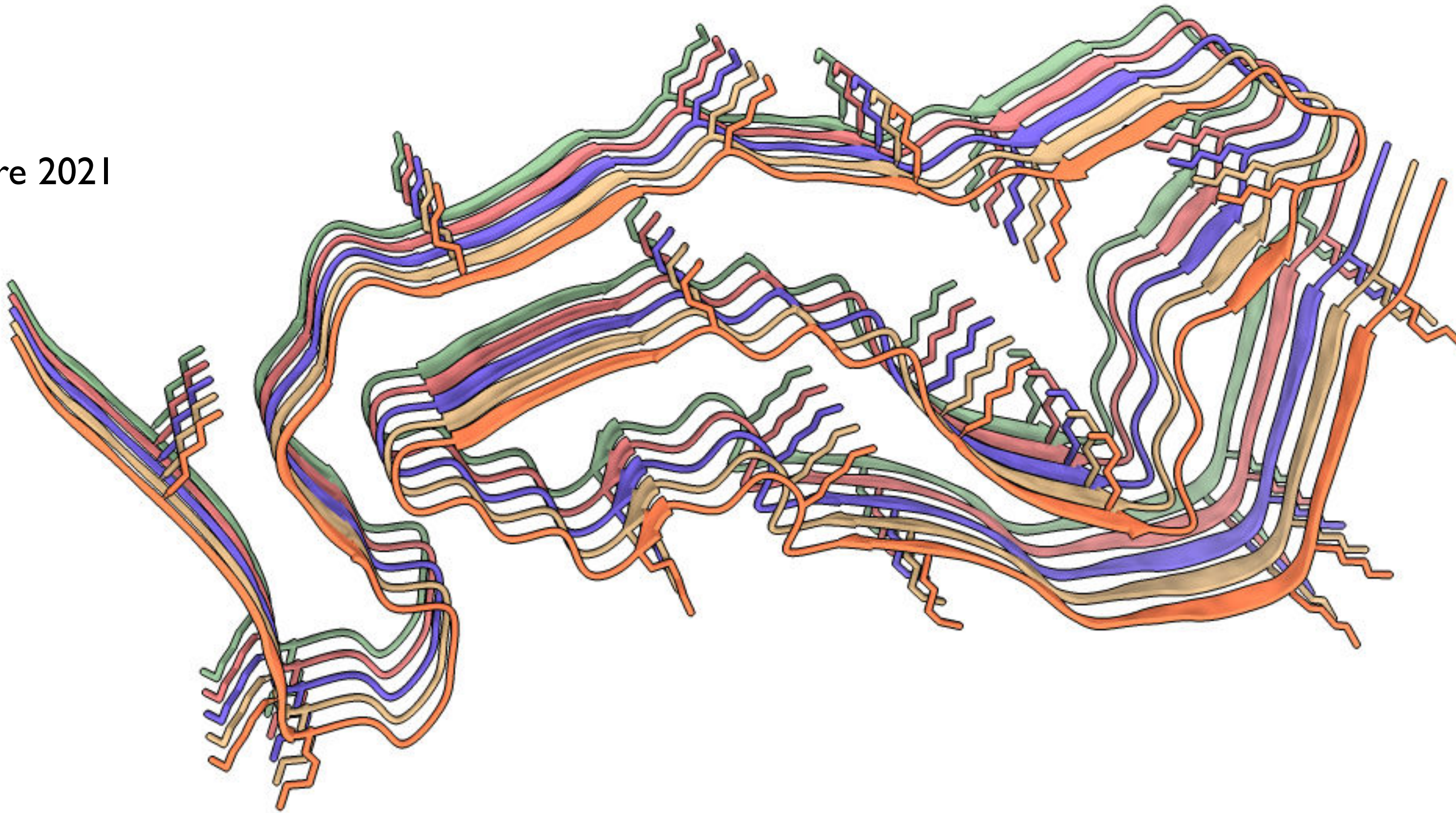


Today's system

Tau filament from the brains of individuals with PSP

*PDB 7p6a
EMDB I3223
resolution 1.9 Å

*Shi, ..., Scheres. Nature 2021



Exercise I

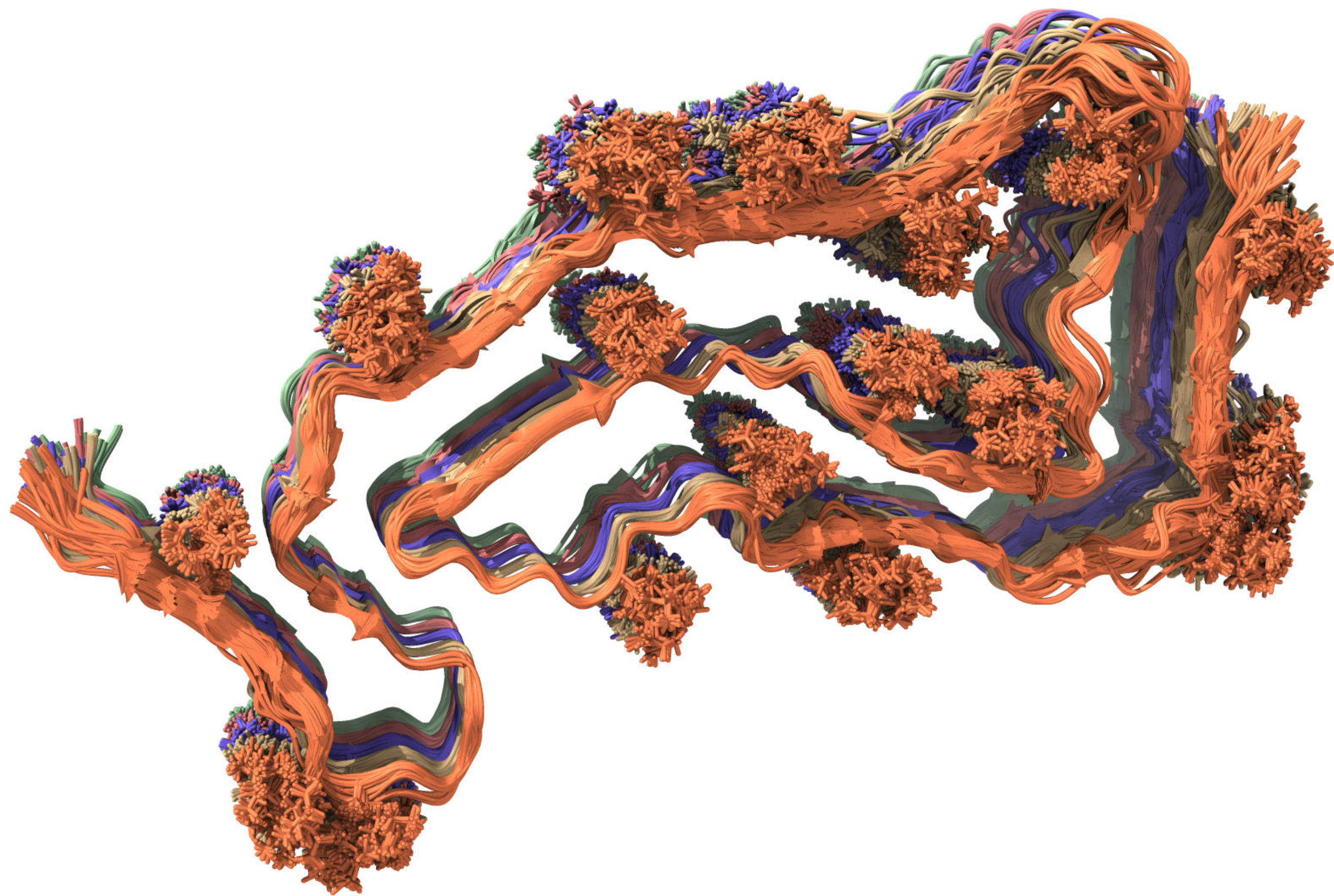
Compare EMMIVox refined model and deposited PDB

		Validation metric	deposited PDB	EMMIVox model
Fit model-map		CC_Mask		
		EMRinger		
Stereochemical quality		Clashscore		
		Molprobability		



Exercise 2

Determining structural ensembles with EMMIVox and Metainference



Fit
model-map

Validation metric	deposited PDB	EMMIVox model	EMMIVox ensemble
CC_Mask			

Acknowledgments



Samuel Hoff
Riccardo Pellarin
Michael Nilges

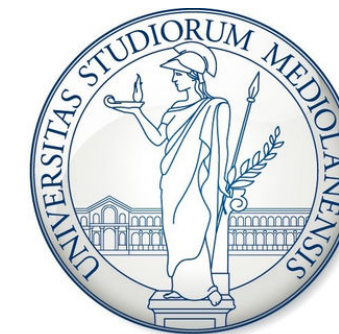


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Scuola
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Superiore di
Studi Avanzati

Giovanni Bussi



Gareth Tribello



Carlo Camilloni

All of you for your attention

Funding



CSCS
Centro Svizzero di Calcolo Scientifico
Swiss National Supercomputing Centre

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